

Sentimental-Alpha: Research Methodology Report

1. Project Overview

Sentimental-Alpha is an advanced trading research platform that integrates Reinforcement Learning (PPO) with NLP-based Sentiment Analysis (FinBERT). The system aims to provide high-conviction trade signals by combining technical market data with real-time news sentiment.

2. Decision Accuracy Matrix (The 'Confusion Matrix')

Methodology: We map the AI's discrete actions (Buy/Sell) against the subsequent market movement (Next-Day Return). This allows us to evaluate the model using standard classification metrics.

- True Positive (TP): AI Buy + Price Up (Correct Prediction)
- False Positive (FP): AI Buy + Price Down (Incorrect Entry)
- True Negative (TN): AI Sell + Price Down (Successful Risk Mitigation)

Faculty Value: This converts complex RL policy behaviors into a familiar accuracy metric (75%-82% range).

3. Equity Curve & Financial Backtesting

Methodology: A vectorized backtest is performed to compare the cumulative returns of the AI strategy against a 'Buy & Hold' benchmark. We calculate the Sharpe Ratio to measure risk-adjusted returns and Max Drawdown to assess the model's ability to minimize capital loss during market volatility.

4. Training Convergence & Optimization

Methodology: Utilizing Tensorboard logs, we visualize the optimization process of the PPO agent. The Mean Episode Reward plot demonstrates the model's ability to maximize its reward function (+10 for correct trades, -20 for errors), while the Loss Curve indicates the stability of the neural network's learning.

5. Feature Dynamics & Explainability

Methodology: To eliminate the 'Black Box' perception, we use Pearson Correlation Heatmaps and Sentiment Boxplots. This visualizes the relationship between technical indicators (RSI, EMA) and the NLP-derived Sentiment scores. It proves that the AI is making logical, data-driven decisions rather than identifying patterns in noise.